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Differences among eighteen winter pea genotypes for forage and cover crop use in the southeastern United States

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Abstract

Winter pea (*Pisum sativum* L.) can be used as a forage and cover crop in the southeast and mid-Atlantic United states; however, minimal effort has been devoted to optimize winter pea genetics for forage and cover crop production in these regions. Studies were conducted from 2015–2017 in Maryland and North Carolina screening 18 winter pea genotypes for forage and cover crop use. Winter pea genotypes were compared with widely grown crimson clover (*Trifolium incarnatum* L.) and hairy vetch (*Vicia villosa* Roth]. All legume genotypes were harvested across four timings. Legume winter hardiness, disease incidence, biomass production, quality, and N release were estimated. Winter hardiness was severe with many winter pea genotypes at the Maryland environments, which restricted winter pea biomass production. There was considerable variation for disease incidence among the winter pea genotypes depending on biotic stressors at each environment. At the North Carolina environments, several winter pea genotypes produced similar biomass to crimson clover and hairy vetch across harvest timings. At the Maryland environments, crimson clover and hairy vetch biomass exceeded winter pea biomass. The winter pea genotypes varied considerably for quality traits including protein, lignin, and cellulose. Relative forage value declined as biomass harvest was delayed and was generally higher with all winter pea genotypes than crimson clover or hairy vetch. These results show wide genetic variation in the winter pea genotypes screened for biomass and quality; this variation could be utilized in breeding efforts to enhance winter pea production in the region.

1 | INTRODUCTION

Cover crop adoption is increasing in the U.S. southeast region, with some states having surpassed 10% of cropland acreage planted with cover crops (USDA–National Agricultural Statistics Service, 2016; Wayman et al., 2016). Farmers are using cover crops for a multitude of reasons including improvement of soil quality, soil moisture conservation, weed suppression, insect suppression, and fertility provision to the subsequent crop (Grandy, Robertson, & Thelen, 2006; Mitchell et al., 2012; Poffenbarger et al., 2015; Schipanski et al., 2014; Vann, Poffenbarger, Zinati, Moyer, & Mirsky, 2017). Legume cover crops are often used for their ability to provide substantial nitrogen (N) to the subsequent crop (Foote, Edmisten, Wells, Jordan, & Fisher, 2014; Parr,

Abbreviations: ADF, acid detergent fiber; NDF, neutral detergent fiber; NIRS, near-infrared spectroscopy; RFV, relative forage value.

Grossman, Reberg-Horton, Brinton, & Crozier, 2011). Legumes typically have low C/N ratios, resulting in rapid decomposition of residue and subsequently allow for fertility provision to the next crop (Wagger, 1989).

Crimson clover (*Trifolium incarnatum* L.) and hairy vetch (*Vicia villosa* Roth) are both frequently used legume cover crops in the southeastern United. Hairy vetch can produce >4,000 kg ha^{-1} dry biomass in the U.S. southeast (Mischler, Duiker, Curran, & Wilson, 2010; Parr et al., 2011) but also produces hard seed, which can create weed management challenges in the subsequent winter cash crops (Maul, Mirsky, Emche, & Devine, 2011; Vann, Reberg-Horton, & Brinton, 2016). Crimson clover can produce substantial biomass and is popular with many farmers in the U.S. southeast but can have notably slower N release than other legume cover crops and has even resulted in N immobilization under some circumstances (Reberg-Horton et al., 2012). Crimson clover growth is also sensitive to wet conditions (Hoveland & Webster, 1965), which often occur over the winter months in the U.S. southeast. In previous research from North Carolina, hairy vetch produced 122–217 kg N ha^{-1} and crimson clover produced 73–156 kg N ha^{-1} (Parr et al., 2011). Winter pea may be another viable winter legume cover crop in this region.

In North Carolina, winter pea not adapted to the U.S. southeast region produced 70–91 kg N ha^{-1} in the Coastal Plain and 128–208 kg N ha^{-1} in the Piedmont regions (Parr et al., 2011). Additional research conducted in the U.S. southeast showed that winter pea could produce more dry matter than hairy vetch (Ranells & Wagger, 1997). Beyond biomass and N production, additional benefits of winter pea include easy termination, no hard seed, and more robust performance across soil types (Sattell, Dick, Hemphill, & McGarth, 1998). In addition, winter pea also has advantages for use as a legume forage crop. Winter pea has a high forage nutritive value with 18–30% crude protein, 35–50% starch, and high concentrations of the amino acid lysine, which are deficient in small grains (McPhee et al., 2003). Winter pea can be used for grazing, cut for hay, or ensiled (Caddel & Enis, 2015; Uzun, Bilgili, Sincik, Filya, & Acikgoz, 2005) and it has potential to replace barley (*Hordeum vulgare* L.) or alfalfa (*Medicago sativa* L.) silage as a forage source for dairy cows in early lactation (Mustafa, Christensen, & McKinnon, 2000).

There are several important requirements for winter pea to succeed in the southeastern and mid-Atlantic regions of the United States as a forage and cover crop; of utmost importance is high biomass production, ideally in excess of 3,000 kg ha^{-1} and comparable to what crimson clover and hairy vetch can produce for biomass in high-yielding situations in these regions (Vann, Reberg-Horton, Castillo, McGee, & Mirsky, 2019). The higher the biomass produced, the more N can be mineralized and available for the subsequent crop. Previous research in the U.S. Pacific Northwest screened multiple winter pea entries in a grain yield trial and reported winter pea biomass ranging from 2300–4760 kg ha^{-1} (McPhee & Muehlbauer, 1999); similar winter pea biomass studies have not occurred in the U.S. southeast. Winter pea is susceptible to a range of abiotic stresses and is most sensitive to cold injury, excessive drought, and heat during flowering (McGee, Nelson, & McPhee, 2013). Winter hardiness in winter pea is complex and controlled by several mechanisms that breeding efforts have targeted in recent years to improve winter hardiness in released cultivars (McGee, Eigenbrode, Newlson, & Schillinger, 2017). Winter pea is also susceptible to a wide range of biotic stresses (McPhee, 2003). Significant efforts have been devoted to incorporating disease resistance in winter pea germplasm (Cousin, 1997), but these efforts have focused on addressing diseases commonly encountered in large winter pea producing regions of the United States. The presence and severity of winter pea diseases are likely different in the U.S. southeast and other U.S. regions where winter pea is bred and produced. Several diseases can restrict winter pea growth, including Fusarium rot root, white mold (*Sclerotinia*), and *Ascochyta* blight, which are often intensified in cool, wet conditions encountered in the winter months in North Carolina

A regional evaluation of winter pea genotypes has not occurred in the U.S. southeast, and growers are limited to using untested cultivars that have been developed in other regions of the United States. Lack of regionally adapted cultivars has been identified as a large barrier limiting cover crop adoption (Wayman et al., 2016). The objectives of this research were to screen winter pea genotypes and compare commonly used legume cover crops in the region, crimson clover and hairy vetch, for their potential use as a forage and cover crop in the U.S. southeast and mid-Atlantic regions. In addition, we seek to provide information on parental selection for future crosses, with the goal of developing regionally adapted winter pea cultivars.

Core Ideas

- There was considerable variation for winter hardiness and disease among the winter pea genotypes
- In North Carolina, several winter pea genotypes produced similar biomass to crimson clover and hairy vetch
- Relative forage value was higher with the winter pea than with crimson clover or hairy vetch

TABLE 1 Soil characteristics by each environment

Environment	Soil series	Textural classification	Soil subgroup
Beltsville north 2016	Hammonton	Loamy sand	Typic Hapludults
Beltsville south 2016	Hatboro	Silt loam	Typic Endoaquepts
Clayton 2016	Varina	Loamy sand	Typic Paleudults
Kinston 2016	Johns	Sandy loam	Typic Hapludults
Clayton 2017	Norfolk	Loamy sand	Typic Kandiudults
Rocky Mount 2017	Goldsboro	Fine sandy loam	Typic Paleudults
Salisbury 2017	Lloyd	Clay loam	Typic Kanhapludults

TABLE 2 Dates for planting and biomass harvest

	Management Event				
Environment	Planting	Harvest 1	Harvest 2	Harvest 3	Harvest 4
Beltsville north 2016[a]	16 Sept.	24 May	7 June	–	–
Beltsville south 2016	16 Sept.	2 May	16 May	31 May	13 June
Clayton 2016	16 Oct.	8 Apr.	22 Apr.	6 May	19 May
Kinston 2016	15 Oct.	8 Apr.	22 Apr.	6 May	19 May
Clayton 2017	29 Oct.	7 Apr.	21 Apr.	3 May	17 May
Rocky Mount 2017	3 Oct.	7 Apr.	21 Apr.	3 May	17 May
Salisbury 2017	23 Oct.	13 Apr.	27 Apr.	9 May	22 May

[a]At the Beltsville north environment, harvest was not possible at the third and fourth timing events because of poor winter pea survival.

2 | MATERIALS AND METHODS

2.1 | Experiment sites and management

Experiments were conducted during the 2015–2016 growing season at two sites on the USDA–ARS Beltsville Agricultural Research Center in Beltsville, MD (North: 39.03˚ N, 76.93˚ W; South: 39.02˚ N, 76.94˚ W); the Central Crops Research Station in Clayton, NC (35.67˚ N,78.51˚ W); and the Caswell Research Farm in Kinston, NC (35.16˚ N, 77.36˚ W); and during the 2016–2017 growing season again in Clayton, at the Upper Coastal Plain Research Station in Rocky Mount, NC (35.90˚ N, 77.68˚ W); and the Piedmont Research Station in Salisbury, NC (35.41˚ N, 80.37˚ W). The year the trial was conducted is defined as the spring when biomass was harvested. The combination of year and location constituted an environment. Soil characteristics for each environment can be found in Table 1. Phosphorus and potassium were applied based on recommendations from state soil testing laboratories. No N was applied and no herbicides were used. Specific management dates for each environment are presented in Table 2.

2.2 | Treatment description

Treatments were 18 winter pea genotypes, one crimson clover cultivar, and one hairy vetch cultivar. Treatments were randomly assigned to experimental units in a randomized complete block design with three to four replications per environment. Seventeen winter pea genotypes were obtained from the USDA–ARS winter pea breeding program at Pullman, WA. Eleven of these genotypes were released winter pea cultivars (CAH-11, Chelan, Common, Fenn, Glacier, Granger, Lynx, Melrose, Romack, Specter, and Windham) and six were advanced ARS breeding lines (PS07300136W, PS09300095W, PS10300120W, PS10300121W, PS06300028W, and PS0017018W). The current winter pea genotype readily available in North Carolina ('Nature's Choice') was also screened. Morphological characteristics of each winter pea genotype are presented in Table 3. The crimson clover cultivar AU Robin and the hairy vetch cultivar Purple Bounty were included as a check comparison since these legume species are used frequently as winter legume cover crops in Maryland and North Carolina. The legume species were inoculated with the appropriate rhizobium species. Winter pea seeds were planted at 444,600 seeds ha^{-1}. Winter pea seed size can vary considerably between genotypes (Miller et al., 2005), therefore winter pea genotypes were seeded using target population densities as opposed to

TABLE 3 Attributes for each winter pea genotype

Genotype	Characteristics					
	Year ABL[a]/released[b]	Growth habit	Flower color	Pigmented seed coat	Leaf type	Flower timing
'CAH-11'	NA	Medium	Purple	Yes	Normal	Medium
'Chelan'	2003/2012	Tall	White	No	Semi-leafless	Medium
'Common'	1940	Tall	Purple	Yes	Normal	Early
'Fenn'	1971	Medium	Purple	Yes	Normal	Late
'Glacier'	1980/1983	Short	Purple	Yes	Normal	Late
'Granger'	1990/1997	Tall	Purple	Yes	Semi-leafless	Medium
'Lynx'	2005	Short	White	No	Semi-leafless	Late
'Melrose'	1973/1978	Tall	Purple	Yes	Semi-leafless	Medium
'Nature's Choice'	2000s	Tall	Purple	Yes	Semi-leafless	Late
PS0017018W	2000	Medium	White	No	Normal	Early
PS06300028W	2006	Medium	White	No	Semi-leafless	Late
PS07300136W	2007	Short	White	No	Semi-leafless	Medium
PS09300095W	2009	Tall	Purple	Yes	Semi-leafless	Early
PS10300120W	2010	Tall	Purple	Yes	Normal	Early
PS10300121W	2010	Medium	Purple	Yes	Normal	Late
'Romack'	1960	Tall	Purple	Yes	Normal	Early
'Specter'	1998/2007	Tall	White	Ghost[c]	Semi-leafless	Medium
'Windham'	1998/2007	Medium	White	Ghost	Semi-leafless	Early

[a] ABL, advanced breeding line.
[b] Cultivar release dates were reported to the best of the authors knowledge. Cultivar release date was not attainable for cv. CAH-11 and Racer.
[c] Ghost mottling is very faint brownish streaks in the testa and is cultivar specific for both Specter and Windham.

seed weight. Crimson clover and hairy vetch were planted at 22.5 kg ha^{-1}. Seeding rates were selected based on average cover crop seeding rate recommendations in the region (Sustainable Agriculture Research and Education, 2007). Plots were established using a cone planter (Hege 1000 series, Wintersteiger Inc.) with row spacing at 17 cm. Plots were planted from mid-September through mid-October (Table 1). Plot size was 1.5 by 2.5 m.

2.3 | Response variables

2.3.1 | Cold injury

Winter pea cold injury (as a measure of winter hardiness) was assessed visually in January and March at the North Carolina environments and in March at the Maryland environments using a 0-to-5 scale, with 0 corresponding to complete death from cold and 5 corresponding to no visual cold injury. Upon arrival at each environment, if no visual cold injury was present across winter pea genotypes then cold injury was not assessed.

2.3.2 | Disease

At the 2017 environments, winter pea disease was assessed visually on a 0-to-100% scale, with 0% corresponding to no disease and 100% corresponding to total death from disease. Diseases were not rated individually; genotypes were scored for overall disease presence from a multitude of diseases that could restrict biomass production and quality. The most prevalent diseases across these environments included *Ascochyta* leaf blight, powdery mildew (*Erysiphe* sp. unknown), and diseases from the *Sclerotinia* complex. Aphids were active on these winter pea genotypes and were not controlled; aphids are vectors of viral pathogens affecting winter pea (McPhee, 2003).

2.3.3 | Legume flower timing

Legume maturity was determined by beginning flowering; genotypes were numerically ranked in order by flower timing across environments and the winter pea genotypes were then categorized into early, medium, and late maturity groups.

2.3.4 | Biomass production

Legume biomass was collected over four sampling periods beginning in mid-April when the earliest genotype started to flower; subsequent harvests occurred 2, 4, and 6 wk following the initial sampling event (Table 2). At each harvest in each environment, aboveground biomass was harvested from two blocks per sampling event using a 0.5-m^2 quadrat randomly placed within the plot. Plants were cut at 0.5 cm above the soil surface. Samples were collected and dried at 65 ˚C to determine dry matter concentration and to calculate dry biomass yield. Samples were then ground using a Thomas-Willy Laboratory Mill (Arthur H Thomas Company) to pass a 1-mm sieve to homogenize biomass in preparation for laboratory analysis. Nitrogen content was calculated using dry biomass and N concentration.

2.3.5 | Biomass quality

Subsamples of the homogenized biomass (5 g) were packed into circular near-infrared spectroscopy (NIRS) ring cups (Part # IH-0386, FOSS North American). A circular NIRS ring cup was filled three-fourths full with ground samples and covered to secure the sample within the cell. The sample was then scanned on a FOSS XDS NIR system (FOSS North America) in reflectance mode, using ISIscan 4.6.11 software and covering both the visible and NIR regions from 400–2498 nm at 2-nm intervals. Each NIR spectrum had a total of 1,050 data points. To avoid contamination, the ring cups were cleaned with a vacuum cleaner between loading the two successive samples. The average spectral properties were used to predict dry-matter, crude protein, fat, nonfibrous carbohydrates, acid detergent fiber (ADF), neutral detergent fiber (NDF), lignin, and ash concentrations. The calibration equation for mixed hay was used (16mh50-2.eqa) for quality analysis of crimson clover and hairy vetch. This was developed by the NIRS Forage and Feed Testing Consortium (http://nirsconsortium.org) and is recommended for use when a cover crop is a legume or a cover crop mixture is >40% legume. Following the initial NIRS analysis, a subset of the winter pea samples was analyzed using wet chemistry to develop a winter-pea-specific NIRS calibration equation; this recently developed calibration equation specific for winter pea was used for analysis of the winter pea genotypes (Saha et al., 2018).

Forage quality was determined by estimating relative forage value (RFV). The RFV is used to rank forages relative to the nutritive value of alfalfa harvested at full bloom, which has a RFV value of 100 (Starks, Brown, Turner, & Venuto, 2015) and is used as an indicator of forage quality (Holman, Thompson, Hale, & Schlegel, 2010). The RFV was calculated using NIRS predicted parameters and the following equation: RFV = DDM (% of dry matter) × DMI (% of body weight)/1.29, where DDM is digestible dry matter, DMI is dry matter intake, and both estimates are calculated using ADF and NDF values previously determined with NIRS (Saha, Hancock, & Kissel, 2017).

TABLE 4 Broad-sense heritability (h^2) among winter pea genotypes estimated on an entry–mean basis in a combined analysis of all environments

Dependent variable[a]	Broad-sense heritability Harvest 1	Harvest 2	Harvest 3	Harvest 4
	%			
Winter hardiness (January)	4	–	–	–
Winter hardiness (March)	52	–	–	–
Disease (harvest)	6	49	74	82
Biomass	61	59	60	50
N content	37	40	42	28
Crude protein	87	85	87	76
Lignin	80	82	21	47
Acid detergent fiber	85	85	81	76
Neutral detergent fiber	85	82	84	71
Relative forage value	84	84	77	82

[a]Winter hardiness was visually estimated once in January (four environments) and once in March (six environments). Disease was estimated at harvest in the North Carolina environments.

Nitrogen release was estimated for the crimson clover AU Robin, hairy vetch Purple Bounty, and the winter pea Romack using the model developed by the University of Georgia College of Agricultural & Environmental Sciences (2020) set to 35 ˚C and optimum soil conditions for decomposition.

2.4 | Statistical analysis

Statistical analyses were conducted using PROC Mixed, Glimix, and PROC Reg in SAS 9.3 (SAS Institute). Data met model assumptions. Legume genotype was a fixed factor; replication and environment were considered random factors. For N release, cold injury, and disease, legume genotype response was presented by individual environment to reflect different abiotic and biotic conditions at each environment. Legume biomass and N content were combined over the North Carolina environments, and the Maryland environments were removed from the combined analysis because of poor winter pea winter hardiness and subsequently poor biomass production at the Maryland environments. All other quality parameters were analyzed in a combined analysis of all environments because of a small variance component for the genotype-by-environment interaction relative to the genotype affect.

Broad-sense heritability (h^2) was estimated for the winter pea genotypes on an entry mean basis using the following equation: $h^2 = \sigma^2_G/(\sigma^2_G + [\sigma^2_{GE}/E] + [\sigma^2_e/RE])$, where σ^2_G is the genetic variance, σ^2_{GE} is the genotype-by-environment variance, σ^2_e is experimental error, R is the number of replications, and E is the number of test environments (Fehr, 1993). Heritability estimates for each dependent variable are presented in Table 4.

Contrasts were made to test linear, quadratic, and cubic functions to find the best fit for the data for N release. The best-fit model was linear and the interaction of species and harvest time was not significant. Plotting of the final results was made using R Software and calculated at 95% confidence intervals.

3 | RESULTS AND DISCUSSION

3.1 | Winter hardiness

Legume genotypes were variably impacted by cold injury at four environments in North Carolina in January (Table 5). By March, there was not visible cold injury with any legume genotypes at the Rocky Mount 2017 and Salisbury 2017 environments. The most severe cold injury in North Carolina was experienced in March at the Kinston 2016 environment, where several winter pea genotypes expressed moderate cold injury (Table 5). Cultivars CAH-11 and Fenn both appeared more winter hardy than the other winter pea genotypes and may be sources of winter hardiness genes. All winter pea genotypes visually appeared to recover from cold injury by April; however, there were slight correlations between cold injury and biomass production at the North Carolina environment. As legume winter hardiness increased, biomass production generally increased (data not shown). Generally, crimson clover and hairy vetch had the same or better winter hardiness than the best winter pea genotype at the North Carolina environments (Table 5). Cold injury among winter pea genotypes was more severe at the northernmost environments in Maryland. Romack was severely injured by cold in both Maryland

TABLE 5 Legume cold injury (0-to-5 scale, 0 corresponding to complete death from cold and 5 corresponding to no visual cold injury; values in bold type) in January and March at individual environments

	Environment									
	Winter hardiness (January)				Winter hardiness (March)					
Genotype	Clayton 2016	Clayton 2017	Kinston 2016	Rocky Mount 2017	Beltsville N 2016	Beltsville S 2016	Clayton 2016	Clayton 2017	Kinston 2016	Salisbury 2017
'CAH-11'	**4.5**	**4.7**	**4.8**	**4.6**	**3.0**	**3.3**	**4.7**	**5.0**	**4.7**	**5.0**
'Chelan'	**4.2**	**4.5**	**5.0**	**4.4**	**1.7**	**2.2**	**4.5**	**5.0**	**2.7**	**5.0**
'Common'	**4.7**	**5.0**	**5.0**	**4.6**	**2.7**	**3.7**	**4.8**	**5.0**	**2.2**	**5.0**
'Fenn'	**4.8**	**4.8**	**5.0**	**4.8**	**2.5**	**2.7**	**5.0**	**4.9**	**4.7**	**5.0**
'Glacier'	**4.5**	**5.0**	**2.8**	**4.5**	**3.2**	**3.2**	**4.3**	**5.0**	**1.7**	**5.0**
'Granger'	**4.5**	**4.4**	**4.7**	**4.4**	**1.7**	**2.5**	**3.8**	**5.0**	**2.3**	**5.0**
'Lynx'	**5.0**	**4.9**	**4.7**	**4.9**	**3.7**	**3.2**	**4.2**	**5.0**	**2.1**	**5.0**
'Melrose'	**4.7**	**4.3**	**4.7**	**4.1**	**1.2**	**2.7**	**3.8**	**4.5**	**2.8**	**4.8**
'Nature's Choice'	**4.7**	**4.8**	**5.0**	**4.6**	**2.7**	**3.3**	**4.3**	**5.0**	**4.0**	**5.0**
PS0017018W	**4.8**	**4.5**	**5.0**	**4.9**	**2.5**	**2.8**	**4.5**	**5.0**	**4.3**	**4.5**
PS06300028W	**4.8**	**4.5**	**4.7**	**3.6**	**1.7**	**2.0**	**4.0**	**4.8**	**4.0**	**5.0**
PS07300136W	**4.8**	**4.5**	**4.7**	**4.5**	**2.0**	**1.6**	**2.7**	**4.0**	**3.0**	**4.4**
PS09300095W	**4.3**	**4.3**	**5.0**	**4.1**	**1.3**	**2.7**	**4.0**	**4.9**	**4.0**	**5.0**
PS10300120W	**5.0**	**4.6**	**5.0**	**4.0**	**1.2**	**1.5**	**4.5**	**5.0**	**3.1**	**4.3**
PS10300121W	**4.7**	**3.4**	**5.0**	**4.1**	**1.5**	**2.3**	**4.2**	**5.0**	**4.0**	**5.0**
'Romack'	**4.8**	**4.5**	**4.7**	**3.8**	**0.5**	**1.7**	**4.5**	**5.0**	**2.1**	**4.8**
'Specter'	**4.8**	**4.6**	**5.0**	**4.5**	**1.8**	**2.5**	**4.2**	**5.0**	**2.7**	**5.0**
'Windham'	**4.7**	**4.7**	**5.0**	**4.9**	**3.2**	**2.8**	**4.0**	**5.0**	**3.3**	**4.8**
Crimson clover[a]	**4.2**	–	**4.7**	**5.0**	**3.3**	**3.7**	**5.0**	–	**5.0**	**5.0**
Hairy vetch	**4.7**	**4.5**	**5.0**	**4.9**	**4.5**	**3.5**	**5.0**	**5.0**	**5.0**	**5.0**
P-value	.04	.003	<.001	<.001	<.001	<.001	.03	.03	.002	NS[b]
LSD (.05)[c]	0.16	0.13	0.22	0.14	0.31	0.26	0.36	0.15	0.59	–

(Continues)

TABLE 5 (Continued)

	Environment									
	Winter hardiness (January)				Winter hardiness (March)					
Genotype	Clayton 2016	Clayton 2017	Kinston 2016	Rocky Mount 2017	Beltsville N 2016	Beltsville S 2016	Clayton 2016	Clayton 2017	Kinston 2016	Salisbury 2017
Average temperature 30 d before rating (˚C)[d]	6.3	7.4	10.1	7.2	4.4	5.0	12.6	11.25	13.2	9.2
Average min. temperature 30 d before rating (˚C)[e]	2.2	3.1	5.6	2.4	−0.5	0.0	6.9	4.7	3.0	1.4
Total precipitation 30 d before rating (cm)[f]	12.3	11.2	11.9	10.3	9.0	9.0	2.7	5.8	4.1	5.1

[a]Data not presented (–) because crimson clover did not survive at this environment.
[b]NS, nonsignificant at the .05 level.
[c]Means are compared within each environment and were separated using Fisher's Protected LSD.
[d]Average temperature 30 d before rating where cold injury was not visible across legume genotypes: Salisbury 2017, 7.5 ˚C; Rocky Mount 2017, 10.7 ˚C.
[e]Average min. temperature 30 d before rating where cold injury was not visible across legume genotypes: Salisbury 2017, 1.8 ˚C; Rocky Mount 2017, 3.1 ˚C.
[f]Total precipitation 30 d before rating where cold injury was not visible across legume genotypes: Salisbury 2017, 14 cm; Rocky Mount 2017, 8.1 cm.

environments (Table 5). Romack has reduced winter hardiness vs. other winter pea genotypes (Auld, Ditterline, Murray, & Swensen, 1983). Unlike the North Carolina environments, where cold injury appeared transient, many winter pea genotypes in Maryland did not recover from severe cold injury. Lynx is a winter pea cultivar that was bred to have improved winter hardiness (McGee et al., 2013); across environments, Lynx generally expressed less cold injury than many other winter pea genotypes (Table 5). Cold injury and biomass production were significantly correlated; as winter hardiness improved there was a linear increase in biomass production at the Maryland environments (data not shown). All winter pea genotypes were more susceptible to cold injury than crimson clover and hairy vetch in Maryland (Table 5). Quantitative trait loci associated with winter hardiness in winter pea have been identified (Dumont et al., 2009); breeding efforts to improve winter hardiness of winter pea germplasm for use in Maryland are underway and necessary for the success of this species in this region (Mirsky, unpublished data, 2019).

3.2 | Diseases

As previously stated, the most prevalent diseases across these environments included *Ascochyta* leaf blight, powdery mildew (*Erysiphe* sp. unknown), and diseases from the *Sclerotinia* complex. Legume genotype response to disease varied by environment (Table 6). Across environments, Glacier, Granger, Melrose, PS09300095W, and Specter generally had the highest disease incidence (Table 6). Previous work in North Carolina screening the grain potential of winter pea also found high disease incidence in Glacier, Granger, Melrose, and PS09300095W (Vann, Reberg-Horton, Castillo, Mirsky, & McGee, 2018). Results were variable, but PS07300136W and PS10300121W generally had lower disease incidence than other winter pea genotypes and may be sources of partial disease resistance. Crimson clover and hairy vetch were less susceptible to disease than any winter pea genotype. At Harvest 1, when disease pressure was generally less severe than later harvests, winter pea biomass production and disease were not correlated (P = not significant). Winter pea biomass production and disease were not correlated at Harvest 4 (P = not significant). As harvest was delayed, winter pea biomass production declined with increasing winter pea disease incidence (Harvest 2: $P = .01$, $Y = 28.6{-}0.002x$, $R^2 = .02$; Harvest 3: $P < .001$, $Y = 50.9{-}0.1x$, $R^2 = .24$). The effect of disease on quality was variable. There was no consistent trend in winter pea N concentration as affected by disease incidence (Harvest 1: P = not significant; Harvest 2: $P = .01$, $Y = 52.3{-}11.5x$, $R^2 = .27$; Harvest 3: P = not significant; Harvest 4: P = not significant). Lignin concentrations of the winter pea genotypes generally increased as disease incidence increased (Harvest 1: $P = <.001$, $Y = -11.7{+}4.8x$, $R^2 = .45$; Harvest 2: $P = .002$, $Y = -0.3{+}3.0x$, $R^2 = .25$; Harvest 3: $P = <.001$, $Y = -22.7{+}5.8x$, $R^2 = .31$; Harvest 4: P = not significant). Late-season disease pressure was severe across all winter pea genotypes at the Kinston 2017, Salisbury 2017, and Clayton 2017 environments. It is possible that late-season *Sclerotinia* or *Ascochyta* blight pressure was restricting late-season biomass production at these environments.

3.3 | Legume maturity

There was variation in initial flower timing between the winter pea genotypes. The earliest-flowering winter pea genotypes began flowering in early April; the late-flowering winter pea genotypes began to flower in mid to late April. In some circumstances, disease pressure restricted flower initiation. The crimson clover cultivar used in this experiment was an early flowering crimson clover cultivar that began flowering shortly after the first flowering winter pea genotype. Hairy vetch started flowering after crimson clover and began flowering with the mid flowering winter pea genotypes. In general, most legume genotypes were flowering by Harvest 2 timing as long as disease pressure was not restricting flower initiation.

3.4 | Biomass production

Broad-sense heritability for biomass was .50–.61 (Table 3). This is high for yield, but the broad genetic diversity associated with the evaluated material may explain these heritability estimates (Scully, Wallace, & Viands, 1991). Winter pea biomass production was drastically different at the Maryland environments, where winter pea biomass was restricted by severe cold injury, therefore, the Maryland and North Carolina environments were analyzed separately.

3.4.1 | North Carolina

Legume genotype affected biomass at all harvests (Table 7). A significant genotype-by-environment interaction was present at Harvest 1, 2, and 4, but the size of the interaction was generally small relative to the size of the genotype main effect, therefore, the analysis was pooled over the North Carolina environments. There was a wide range in biomass production among the winter pea genotypes; biomass was 2,007–5,795; 2,241–7,018; 2,191–5,357; and 1,546–5,751 kg ha^{-1} at Harvest 1, 2, 3, and 4, respectively (Table 8). Fenn, Granger, and PS09300095W were high-biomass-producing winter pea genotypes across all harvests. Romack and Chelan had moderate biomass production at early harvests; however, by May they were among the top biomass producers (Table 8). These are not the winter pea genotypes that have been commonly

TABLE 6 Legume disease incidence as affected by genotype at individual environments

Genotype	Environment														
	Clayton 2016			Clayton 2017			Kinston 2016	Rocky Mount 2017				Salisbury 2017			
	22 April	6 May	19 May	7 April	21 April	3 May	6 May	7 April	21 April	3 May	17 May	13 April	27 April	9 May	22 May
	%														
'CAH-11'	7	13	35	23	18	30	35	8	11	10	45	40	57	78	100
'Chelan'	8	10	23	8	25	18	95	0	8	5	33	33	50	50	83
'Common'	10	18	50	17	35	50	100	0	4	5	30	10	42	50	95
'Fenn'	9	8	40	15	22	55	63	3	5	10	30	13	35	70	98
'Glacier'	35	45	55	3	12	45	–	15	23	20	55	13	35	75	100
'Granger'	50	40	85	18	23	53	65	23	38	8	43	10	45	55	98
'Lynx'	3	6	43	28	4	50	95	10	23	8	38	23	45	80	100
'Melrose'	42	30	30	10	22	70	95	5	23	5	60	70	78	83	100
'Nature's Choice'	6	10	25	15	28	30	85	5	10	8	25	30	25	53	100
PS0017018W	2	5	25	13	30	18	85	5	5	5	23	45	58	48	98
PS06300028W	1	5	35	28	38	25	35	0	8	13	35	38	18	60	100
PS07300136W[a]	3	8	15	11	5	15	–	3	3	8	18	20	13	10	90
PS09300095W	18	6	35	15	23	85	75	8	35	5	45	13	15	53	90
PS10300120W	4	8	40	10	13	65	95	10	10	5	30	50	75	45	98
PS10300121W	10	20	45	8	16	25	60	3	10	5	35	10	48	40	100
'Romack'[a]	21	28	35	20	5	30	–	3	20	8	45	48	45	68	100
'Specter'	21	23	80	8	11	83	65	8	38	5	45	63	70	73	100
'Windham'	7	18	10	3	10	45	–	0	5	20	15	55	10	70	100
Crimson clover[a]	0	0	0	–	–	–	0	0	0	0	0	3	0	0	0
Hairy vetch	0	0	0	0	0	0	15	0	0	0	0	0	0	0	0
P-value	.02	NS[b]	.05	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	<.001
LSD (.05)[c]	9	–	14	–	–	–	–	–	–	–	–	–	–	–	4

[a]Data not presented (–) because legume genotype had poor survival at these environments.
[b]NS, non-significant at the .05 level.
[c]Means are compared within each environment and were separated using Fisher's Protected LSD.

TABLE 7 ANOVA for the effect of legume genotype (G), environment (E), and the legume genotype-by-environment (G×E) interaction on biomass and N content in a combined analysis of the North Carolina (NC) and the Maryland (MD) environments

		Biomass				Nitrogen content			
		Harvest timing[a]							
Analyses	Source	One	Two	Three	Four	One	Two	Three	Four
		$P > F$ (mean square)							
NC combined	G	.001 (7,701,713)	<.001 (17,613,191)	.02 (7,378,705)	.05 (5,869,410)	NS[b] (5,968)	.001 (12,873)	NS (4,539)	NS (3,138)
E	.001 (37,146,543)	NS (43,963,794)	.01 (90,088,335)	NS (15,047,415)	.001 (42,742)	NS (43,350)	.003 (100,917)	NS (6,632)	
	G×E	.001 (2,878,634)	.006 (4,939,061)	NS (3,907,683)	.05 (3,381,399)	.002 (3,617)	.03 (4,741)	.02 (3,780)	NS (2,506)
MD combined[c]	G	<.001 (3,770,549)	NS (4,011,469)	NS (2,291,550)	NS (1,711,793)	.001 (2,165)	NS (1,195)	NS (1,236)	NS (1,608)
	E	.02 (5,077,155)	NS (11,267,259)	–	–	NS (4,551)	–	–	–
	G×E	.001 (679,165)	NS (2,285,289)	–	–	NS (377)	–	–	–

[a]Harvest timings are as follows: One: NC 7–13 April, MD 2–24 May; Two: NC 21–27 April, MD 16 May–7 June; Three: NC 3–9 May, MD 31 May; Four: NC 17–22 May, MD 13 June.

[b]NS, non-significant at the .05 level.

[c]Data not presented (–) because the third and fourth harvest events were not possible at the Beltsville North environment and the results are only representative of one environment.

TABLE 8 Legume biomass and N content combined over the North Carolina (NC) and Maryland (MD) environments

Genotype	Biomass								Nitrogen content[a]							
	Combined NC				Combined MD[b]				Combined NC				Combined MD[b]			
	Harvest timing[c]															
	One	Two	Three	Four	One	Two	Three	Four	One	Two	Three	Four	One	Two	Three	Four
	kg ha															
'CAH-11'	4,078	4,158	4,410	3,537	834	1,811	1,858	718	122	117	126	83	27	88	51	19
'Chelan'	3,753	3,636	4,590	5,363	504	514	828	2,580	104	92	118	128	20	27	21	65
'Common'	3,392	4,866	4,279	4,676	885	1,269	1,441	2,128	104	144	127	114	30	54	46	58
'Fenn'	4,182	7,018	5,135	4,214	383	476	1,660	2,265	138	220	152	109	17	32	45	65
'Glacier'	2,669	3,306	2,191	3,126	475	743	803	1,778	99	110	79	120	21	45	23	51
'Granger'	3,994	5,940	4,456	4,870	772	391	770	68	116	170	127	109	50	23	22	7
'Lynx'	2,007	2,241	2,265	1,546	186	184	558	978	68	75	80	60	8	10	19	31
'Melrose'	3,469	3,834	3,260	3,677	688	1,154	916	1,117	98	105	81	85	28	74	20	26
'Nature's Choice'	3,553	4,585	4,625	4,629	1,157	1,363	2,968	447	109	142	129	120	41	73	80	13
PS0017018W	3,720	4,224	5,102	3,505	616	742	2,387	1,917	110	122	139	85	19	40	71	51
PS06300028W	2,074	2,779	3,715	2,788	161	380	339	219	74	91	106	82	7	17	10	6
PS07300136W	2,763	3,033	3,137	3,093	65	2,487	405	900	98	100	110	98	2	5	13	27
PS09300095W	4,514	5,632	4,175	5,295	517	819	967	1,747	138	163	122	130	23	54	29	48
PS10300120W	3,897	4,693	4,656	3,386	215	317	815	973	115	132	124	81	9	20	23	28
PS10300121W	3,806	4,960	4,720	3,466	965	1,272	1,792	1,787	125	147	136	78	36	59	46	45
'Romack'	3,327	4,927	4,587	5,751	361	605	1,492	1,485	97	137	121	159	12	37	40	37
'Specter'	4,159	5,307	3,226	4,728	495	467	575	491	125	147	90	126	19	24	15	11
'Windham'	2,331	2,892	3,324	2,825	219	421	686	396	79	90	99	91	7	28	20	11
Crimson clover	5,795	7,108	4,619	4,911	3,823	3,455	4,081	2,930	153	164	89	100	89	78	85	58
Hairy vetch	4,672	5,975	5,357	5,385	3,153	3,435	3,612	3,270	166	175	149	149	89	80	87	74
P-value	.001	<.001	.02	.05	<.001	NS	NS	NS	NS	.001	NS	NS	.0[illegible]	NS	NS	NS
LSD (.05)[d]	515	677	617	713	380	–	–	–	–	21	–	–	11	–	–	–

[a]N content is calculated via multiplying biomass by N concentration.

[b]MD biomass and N content at harvest timings two, three, and four only represent the Beltsville South environment.

[c]Harvest timings are as follows: One: NC 7–13 April, MD 2–24 May; Two: NC 21–27 April, MD 16 May–7 June; Three: NC 3–9 May, MD 31 May; Four: NC 17–22 May, MD 13 June.

[d]Means were separated using Fisher's protected LSD.

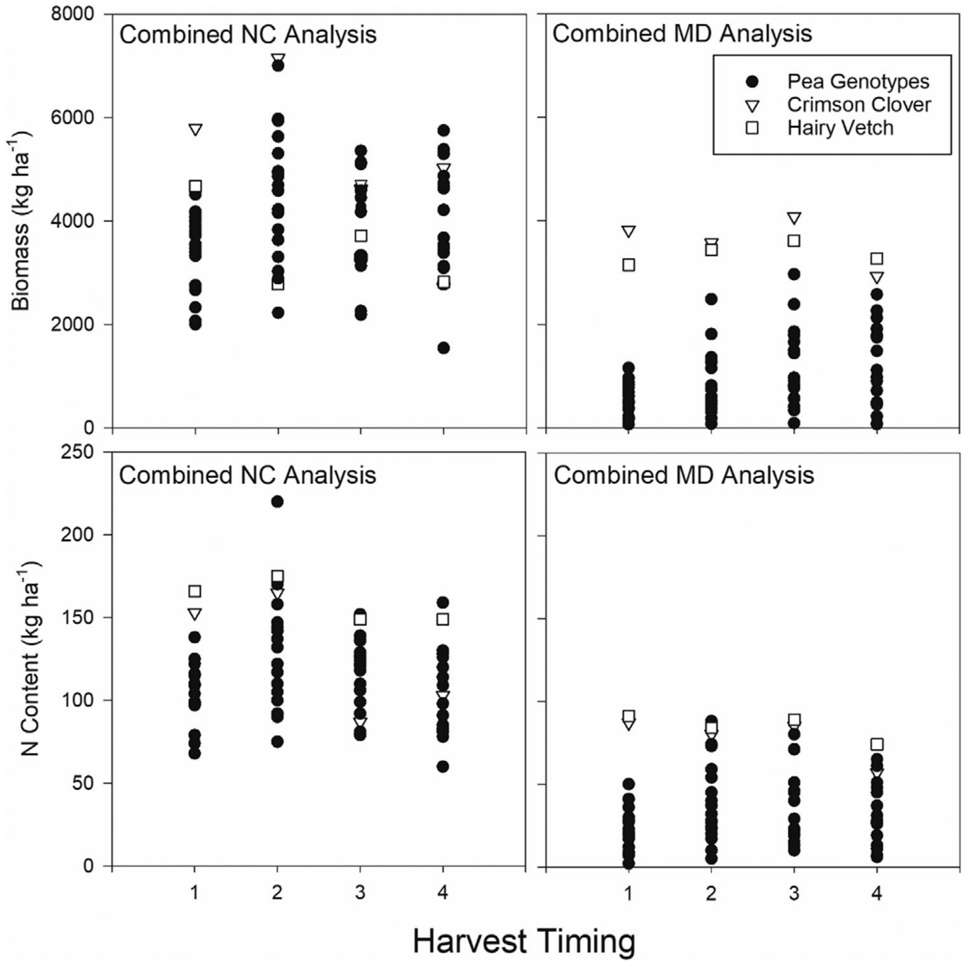

FIGURE 1 Variation in biomass and N content as affected by legume genotype in a combined analysis of the North Carolina environments and the Maryland environments

evaluated for biomass production in the U.S. southeast region (Parr et al., 2011; Ranells & Wagger, 1997); these results indicate that greater winter pea biomass can be achieved than reported from previous research conducted in the region. It is worth noting that Granger produced moderately high biomass (Table 8) despite relatively high disease pressure (Table 6). Lynx was the lowest biomass producing winter pea genotype (Table 8). Lynx is a winter pea cultivar that was bred for improved winter hardiness, palatability, and disease resistance (McGee et al., 2013). There were several winter pea genotypes that produced similar biomass to crimson clover and hairy vetch across harvests (Figure 1), indicating that an adapted winter pea genotype can be competitive with crimson clover and hairy vetch for biomass production.

Leaf type affected winter pea biomass at Harvests 1, 2, and 3 ($P = .007$, $P = .08$, and $P = .02$, respectively); genotypes with normal leaves produced more biomass than did semi-leafless genotypes. These results are unlike previous research that documented no difference in forage yield between winter pea lines of differing leaf type (Uzun et al., 2005). While this general trend existed in the North Carolina environments, there were still several semi-leafless winter pea genotypes that were top biomass producers (Chelan, Granger, and PS09300095W).

Flowering timing did not affect biomass across harvests (P = not significant). Many winter pea genotypes reached maximum biomass around Harvest 2; however, some reached maximum biomass at the final harvest (Chelan and Romack). Most winter pea genotypes had not reached maximum biomass at Harvest 1. PS09300095W and Romack were the earliest-flowering winter pea genotypes, and both had higher biomass at Harvest 4 than Harvest 1 (Table 8). These are likely indeterminate winter pea genotypes capable of simultaneous vegetative and reproductive growth. In North Carolina, there is a wide range of ideal planting dates for summer cash crops {corn (*Zea mays* L.), cotton (*Gossypium* spp.), soybean [*Glycine max* (L.) Merr.], peanut (*Arachis hypogaea* L.)} and specialty crops. Corn producers in North Carolina would be interested in a winter pea cultivar that reached maximum biomass production in mid-April, whereas a cotton or soybean producer might be more interested in a winter pea cultivar that reached maximum biomass in mid-May. Crimson clover had the highest biomass at early harvests and biomass declined as harvest was delayed (Figure 1). An early maturing crimson clover genotype was used in this experiment and may explain the reduction in biomass as harvest was delayed. Hairy vetch biomass remained relatively consistent across harvests (Figure 1).

3.4.2 | Maryland

Legume genotype affected biomass at Harvest 1 (Table 7). Winter pea biomass production was restricted by cold injury over the winter in this region (Table 5). There was a substantial range in biomass production among the winter pea genotypes; biomass was 65–1,157; 317–2,487; 339–2,968; and 68–2,580 kg ha^{-1} at Harvests 1, 2, 3, and 4, respectively (Table 7). There were several winter pea genotypes, including Chelan, Common, Fenn, and PS0017018W, that produced moderate biomass at these environments despite cold injury (Tables 5 and 8). Lynx was one of the most winter hardy winter pea genotypes in this environment but still produced very little biomass (Table 5). This cultivar may be a good source of cold resistance genes but is of limited value for biomass production as was similarly observed in North Carolina. At the Maryland environments, biomass increased as harvest was delayed for many winter pea genotypes (Figure 1). As was observed in North Carolina, Chelan and Romack produced relatively substantial biomass by Harvests 3 and 4 (Figure 1). Crimson clover biomass declined as harvest was delayed in the Maryland environments and hairy vetch biomass remained relatively consistent across harvests (Figure 1), consistent with results from North Carolina. Across harvests, crimson clover and hairy vetch produced more biomass than any winter pea genotype in Maryland, and this was particularly true for the earlier harvests. Crimson clover and hairy vetch are more adapted for production in the Maryland region, and efforts to enhance winter pea winter hardiness will be critical for winter pea to compete or compliment with crimson clover or hairy vetch for forage and cover crop use.

3.5 | Quality

Separate analyses were conducted for the North Carolina and Maryland environments for N content because of the large differences in legume biomass production between the two regions.

3.5.1 | North Carolina N content

There was a wide range in N content among the winter pea genotypes (Figure 1), which is a function of both variability

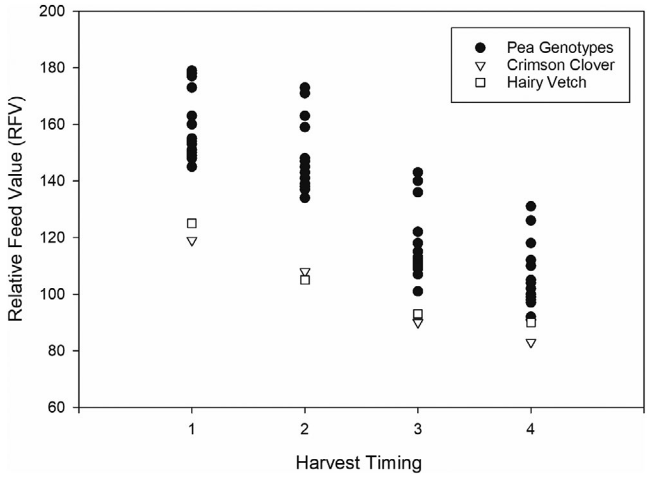

FIGURE 2 Relative forage value (RFV) across harvest timings as affected by legume genotype in a combined analysis of all environments

in biomass production and significant variability in N concentration between the winter pea genotypes. High-N-content-producing winter pea genotypes include PS09300095W, Fenn, Granger, Specter, and PS10300121W (Table 8). At Harvest 1, crimson clover and hairy had higher N content than any winter pea genotype; however, by Harvest 2, 3, and 4 timings, there were winter pea genotypes that were producing as much, or more, N as crimson clover and hairy vetch (Figure 1). As harvest timing was delayed, crimson clover N content was reduced drastically (Figure 2). Hairy vetch consistently produced high N content (Figure 2).

3.5.2 | Maryland N Content

In the Maryland environments, crimson clover and hairy vetch had the highest N content across Harvests 1–3; however, at the final harvest crimson clover, N content declined, as was observed North Carolina (Figure 1). Cold injury resulted in overall low winter pea N production in Maryland, but among the winter pea genotypes, PS10300121W and Nature's Choice consistently produced the most N across harvests.

3.5.3 | Other quality traits

Legume genotype affected many other quality traits (Table 9). A significant genotype-by-environment interaction was often detected for quality traits across harvests (Table 9), but the size of the interaction was generally quite small relative to the size of the genotype main effect, therefore, analysis for quality traits were pooled across all environments. There was a range in crude protein concentration among the winter pea genotypes with 182–228, 167–220, 143–223, and 136–204 g kg^{-1} at Harvests 1, 2, 3, and 4, respectively, and many winter pea genotypes exceeded crimson clover and hairy vetch for crude

TABLE 9 ANOVA for the effect of legume genotype (G), environment (E), and the legume genotype-by-environment (G×E) interaction on various quality parameters in a combined analysis of all environments

Dependent variable	Source	Harvest timing One	Two	Three	Four
		$P > F$ (mean square)			
Crude protein	G	<.001 (3,935)	<.001 (3,931)	<.001 (5,659)	<.001 (2,724)
	E	.001 (6,997)	.007 (3,659)	.001 (7,923)	.04 (2,809)
	G×E	.03 (545)	.02 (466)	.06 (563)	.001 (614)
Lignin	G	<.001 (754)	<.001 (686)	.01 (418)	.002 (412)
	E	<.001 (8,836)	<.001 (9,775)	<.001 (7,250)	<.001 (2,407)
	G×E	<.001 (112)	<.001 (124)	<.001 (207)	<.001 (154)
Acid detergent fiber	G	<.001 (9,715)	<.001 (11,046)	<.001 (15,157)	<.001 (6,398)
	E	<.001 (32,278)	<.001 (44,963)	<.001 (38,596)	.001 (11,379)
	G×E	.04 (603)	.001 (967)	.003 (1,949)	.004 (1,543)
Neutral detergent fiber	G	<.001 (9,850)	<.001 (16,101)	<.001 (12,602)	<.001 (8,860)
	E	.001 (17,575)	<.001 (17,715)	.001 (53,337)	<.001 (73,363)
	G×E	.12 (716)	968 (.001)	.09 (1,324)	<.001 (1,988)
Relative forage value	G	<.001 (3,080)	<.001 (3,441)	<.001 (2,109)	<.001 (935)
	E	.001 (7,320)	<.001 (6,703)	<.001 (4,843)	<.001 (3,959)
	G×E	.08 (263)	.003 (282)	.03 (195)	.04 (147)

protein concentration across harvests (Table 10). Enhancing crude protein intake can be desirable for forage consumption (Carr, Horsley, & Poland, 2004). Lignin concentration was generally higher in hairy vetch and crimson clover than the winter pea genotypes (Table 10). Higher levels of lignin and NDF have been reported to slow residue decomposition by reducing microbial attack on the residue (Ranells & Wagger, 1996; Sievers & Cook, 2018), and this could slow N release from legume cover crops. Legume cover crop N release often occurs prior to optimal corn N uptake (Sievers & Cook, 2018), and selecting a legume cover crop with high lignin and high N content could optimize N delivery to the following cash crop. The ADF and NDF values were higher for crimson clover and hairy vetch than for any winter pea genotype (Table 10). Acid detergent fiber represents the least digestible portion of the forage and, as ADF increases, the digestibility of the forage declines (Saha et al., 2010). Neutral detergent fiber can be used to predict dry matter intake; as NDF increases, forage consumption by the animal decreases (Saha et al., 2010). Lignin, ADF, and NDF all increased as harvest was delayed (Table 10).

Relative forage value was affected by legume genotype across harvests ($P < .001$). All winter pea genotypes had higher RFV than crimson or hairy vetch (Figure 3). Relative forage value declined across all legume genotypes as harvest was delayed (Figure 3). Of the winter pea genotypes, Lynx and Glacier generally had the highest RFV followed by PS06300028W, PS07300136W, and Windham (data not shown).

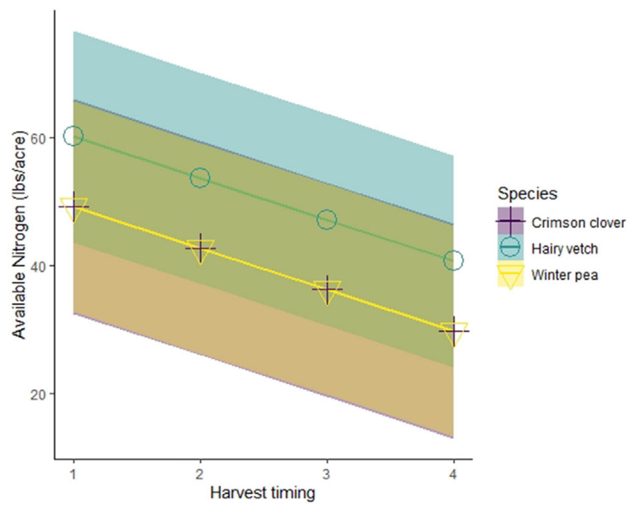

FIGURE 3 Available nitrogen across four different harvest times from the winter pea cultivar Romack compared with crimson clover and hairy vetch. Shading represents a 95% confidence interval

Nitrogen release was estimated for one genotype of each species (Romack [winter pea], AU Robin [crimson cover], and Purple Bounty [hairy vetch]) using four samples collected at 2-wk intervals after the first winter pea plant had flowered. Legume species and sampling period affected N release (P = .0318, P = <.01, respectively) but the interaction of the terms did not (P = .0816), meaning the response across sampling periods for legume species impact on N release was consistent. According to Tukey–Kramer grouping, LSMeans

TABLE 10 Crude protein, lignin, acid detergent fiber (ADF), and neutral detergent fiber (NDF) across biomass harvest timings in a combined analysis of all environments

Genotype	Crude protein				Lignin				ADF[a]				NDF[a]			
	Harvest timing															
	One	Two	Three	Four	One	Two	Three	Four	One	Two	Three	Four	One	Two	Three	Four
	g kg															
'CAH-11'	190.0	179.6	174.3	156.1	64.8	71.6	84.8	91.4	297.2	318.3	381.2	420.5	405.0	408.2	489.4	514.0
'Chelan'	189.1	167.3	161.3	151.6	58.5	71.4	82.2	89.6	287.5	325.7	383.6	435.7	401.5	418.6	503.2	539.5
'Common'	196.1	182.4	184.1	155.7	62.3	72.1	81.4	95.5	289.2	328.3	370.7	438.9	388.5	415.1	476.2	520.1
'Fenn'	203.5	193.9	174.3	158.9	63.1	76.0	85.9	91.2	312.1	336.0	395.0	424.2	404.7	422.0	503.1	511.3
'Glacier'	214.5	198.2	203.2	202.8	58.4	68.8	82.4	87.0	254.2	297.2	316.0	366.6	361.9	379.8	430.7	433.5
'Granger'	181.5	177.9	170.4	145.1	61.9	74.8	87.6	101.5	305.2	335.0	396.1	463.1	409.1	4430.1	498.2	554.9
'Lynx'	226.0	220.7	223.0	204.0	52.6	60.3	85.7	78.2	252.1	278.0	302.4	357.2	366.3	370.4	423.3	451.9
'Melrose'	181.5	173.0	143.3	136.4	63.6	75.1	88.9	93.6	302.7	331.1	418.2	458.7	405.3	418.9	522.7	549.7
'Nature's Choice'	197.3	193.0	176.0	164.0	60.9	72.2	80.1	91.6	294.2	323.4	388.1	414.4	400.4	409.1	492.4	510.2
PS0017018W	187.9	180.8	172.8	156.9	63.0	73.0	85.7	92.0	303.9	326.8	402.2	416.5	401.1	412.2	503.5	507.3
PS06300028W	225.9	207.7	185.0	167.6	48.6	62.4	77.9	80.7	260.7	291.5	375.4	396.2	362.9	380.6	485.3	486.6
PS07300136W	228.0	215.5	216.7	185.4	51.4	58.0	75.3	87.2	267.4	278.0	323.8	372.5	369.2	367.4	441.3	475.7
PS09300095W	189.4	178.2	178.2	155.3	60.8	70.5	84.9	85.3	300.0	325.5	382.1	417.6	410.0	417.7	490.1	516.2
PS10300120W	194.8	186.9	161.5	148.9	66.8	72.7	87.7	92.8	308.3	320.0	402.1	431.8	414.0	405.1	505.7	522.1
PS10300121W	200.0	181.6	172.2	153.0	65.1	74.8	87.9	98.3	307.7	338.8	383.1	430.6	405.6	423.4	496.2	529.7
'Romack'	186.3	170.9	164.7	149.1	66.0	73.7	84.5	90.3	311.0	334.3	386.4	432.0	417.4	434.2	500.0	528.5
'Specter'	186.7	170.0	166.1	157.0	59.7	74.4	86.3	94.2	299.4	335.1	395.7	430.8	406.8	424.7	495.1	506.9
'Windham'	208.1	200.0	187.2	172.6	60.3	65.4	81.0	86.3	276.0	292.4	356.2	401.1	388.2	387.6	469.9	491.1
Crimson clover	160.0	140.0	121.1	123.2	75.0	82.9	98.0	100.1	363.5	393.5	447.4	470.3	471.0	506.9	558.6	581.5
Hairy vetch	203.8	176.9	166.4	161.2	82.7	92.4	101.6	111.7	351.4	395.1	440.7	447.3	459.8	511.0	552.5	563.6
P-value	<.001	<.001	<.001	<.001	<.001	<.001	.01	.002	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
LSD (.05)[b]	6.0	5.7	6.7	9.5	2.7	2.9	4.1	4.6	6.4	8.5	12.3	14.7	6.9	8.2	10.3	16.6

[a]Total cellulose is calculated as the sum of cellulose and hemicellulose.
[b]Means were separated using Fisher's protected LSD.

of N release for hairy vetch (50.0 kg ha^{-1}) was significantly higher than crimson clover (38.7 kg ha^{-1}), but winter pea was positioned in the middle (41.2 kg ha^{-1}) and not significantly different than either. The best-fit model to describe N release was determined to be linear, and since species and sampling time interaction was not significant, the slope was the same for all three species, with different intercept and confidence intervals (Figure 2). The results indicate that winter pea is competitive with both hairy vetch and crimson clover for N release when winter pea hardiness is adequate and disease is not restricting biomass production and that winter pea can be a good cover crop choice to increase N in soil in the North Carolina environments evaluated in this study

Winter hardiness will need to be improved for winter pea to be competitive with crimson clover and hairy vetch for biomass production and N release in the Maryland environments. Winter hardiness had a minimal impact on winter pea biomass production in North Carolina. All winter pea genotypes screened in this experiment were more susceptible to diseases than crimson clover or hairy vetch, and breeding efforts to enhance disease resistance for production in the region is merited. There was considerable variation in biomass production and quality among winter pea genotypes evaluated in this study. In North Carolina, some of the winter pea genotypes screened can produce similar biomass quantities and release as much N as crimson clover and hairy vetch. Many of these winter pea genotypes have higher RFV than crimson clover or hairy vetch and the fact that they can produce similar biomass in North Carolina may make this species more desirable for forage end uses. Relative forage value generally declined as harvest was delayed, but the effects of harvest timing on biomass production was genotype specific. These results demonstrate that genotypic variation among winter pea genotypes can be agronomically important and that the top performing winter pea genotypes are competitive with crimson clover and hairy vetch in environments where winter hardiness does not restrict biomass production. In addition, previous research has found that winter pea may be a better fit for production in mixture with a small grain than hairy vetch because it is not as competitive in mixture thereby allowing the small grain to also produce considerable biomass (Vann et al., 2019). Both cover crop and forage producers often plant species mixtures to ensure they are gaining benefits from both species in the mixture (Vann et al., 2019). Forage quality of many winter pea genotypes is superior to both crimson clover and hairy vetch. While breeding efforts to increase biomass production and N concentration would be valuable, the biomass production of the genotypes currently available are quite good, and more short-term value would be gained from breeding efforts focused on developing winter-hardy winter pea genotypes for production in the northern regions of the U.S. southeast and enhancing disease resistance for pathogens prevalent in the region.

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